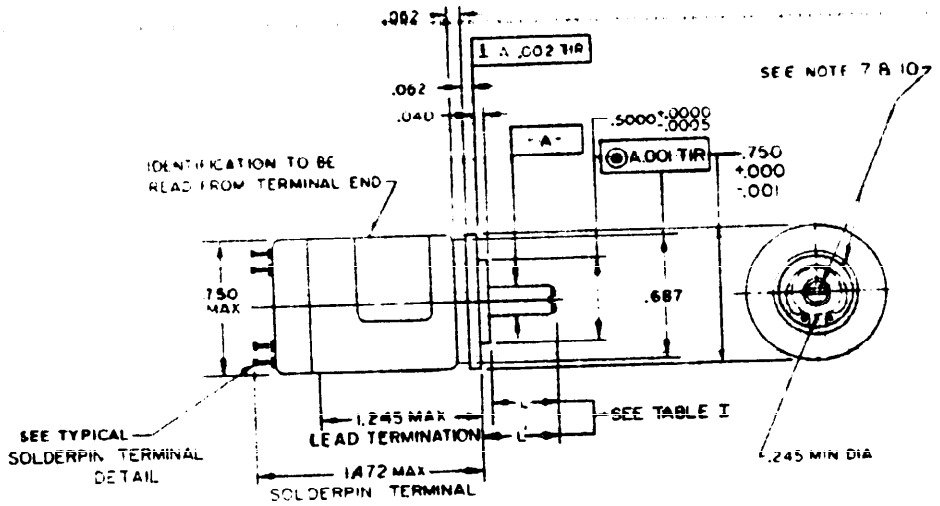
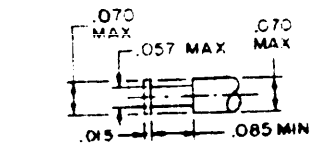
MILITARY SPECIFICATION SHEET

TRANSOLVER
TYPE 11.8 08R37Y440A

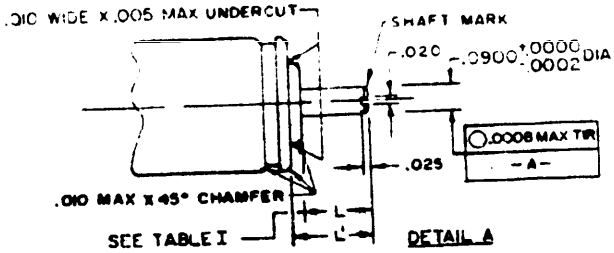
The complete requirements for procuring the transolver described herein shall consist of this document and the latest issue of MIL-T-83727.

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

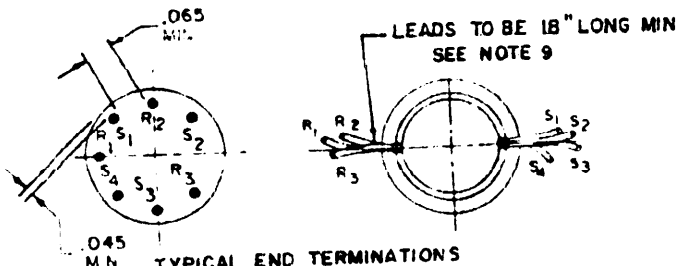




TYPICAL SOLDERPIN TERMINAL DETAIL



DETAIL A



TYPICAL END TERMINATIONS
SEE NOTE 4, 5 & 6

FSC 5990

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NOTES:

1. Dimensions are in inches.
2. Unless otherwise specified, tolerances are ±.005 on decimals, ±2° on angles.
3. Perpendicularity and concentricity shall be measured with the unit in a vertical position supported by the shaft, then the housing rotated.
4. Terminal sequence shall be as shown and identified by molded lettering on the end cap or permanently affixed thereto. Terminal lead wires shall depart through the back surface of the terminal end of the transolver in any acceptable manner.
5. [illegible] and contouration are permitted for undimensioned design detail.
6. [illegible] to terminal shall be soldered with high temperature [illegible] type 60/35.
7. Mark on housing is to match shaft mark within 10° when rotor is set at approx. transolver zero and will bear no relation to the center lines shown on the drawing.
8. Pending issuance of a qualified products list (QPL) a sample of the transolvers to be furnished under this specification shall be subjected to the first article tests as specified in MIL-T-83727.
9. Leads shall be type L-28.
10. The front retainer shall be threaded or a snap ring.

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REQUIREMENTS:	VALUE	UNIT	TOLERANCE
Primary element	Rotor	-	-
Excitation frequency	400	Hz	±1%
Primary voltage	11.8	Volts	±1%
Primary current	30	Milliamperes	Max.
Primary power	.07	Watts	Nominal
D.C. resistance:			
Rotor	99	Ohms	±15%
Stator	378	Ohms	±15%
Impedance:			
Z_{ro}	108+j363	Ohms	R±15%+jx±10%
Z_{so}	545+j1970	Ohms	R±15%+jx±10%
Z_{ss}	780+j120	Ohms	Referenced
Transformation ratio:			
Output input	2.203	-	±2%
Phase shift	12	Degrees	±1.5°
Electrical error	7	Minutes	Max.
Interaxis error	5	Minutes	Max.
Null voltage:			
Fundamental	30	MV rms	Max.
Total	40	MV rms	Max.
Friction torque:			
At 23° C	.04 (3)	Oz-in. (Gm-Cm)	Max.
At -55° C and +125° C	.1 (9)	Oz-in. (Gm-Cm)	Max.
Radial play	.0004 (Using 4 oz. load)	Inches	Min.-Max.
End play	.0001-.0008 (Using 8 oz. load)	Inches	Max.
Temperature rise	5	°C	

Part number: M83727 2 and a dash number from table I.

TABLE I. Dash number and characteristics.

Dash no.	Shaft diameter	Shaft style	Termination	Shaft length	
				L ref	L' dim to first mounting flange surface (±.010)
-001	.0900	Plain	Wire leads	.335	.375
-002	.0900	Plain	Solder pins	.335	.375
-003	.0900	Plain	Wire leads	.500	.540
-004	.0900	Plain	Solder pins	.500	.540

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Custodians:
Army - MU
Navy - AS
Air Force - 80

Review activities:
Army - EL, MU, AV
Navy - SH, EC
Air Force - 11, 17
DSA - ES

User activities:
Navy - CG, MC
Air Force -

Preparing activity:
Air Force - 80

Agent:
DSA - ES

(Project 5990-0258)

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